

THIS IS A TEST FILE

Package **intexgral** Version: 3.0.0 - 2025-12-24

A LaTeX package for typesetting integrals.

Simple test

$$\int x \, dx \quad (1)$$

Key **limits** test

$$\int_a^b x \, dx \quad (2)$$

$$\int_a^b \int_c^d \int_e^f x \, dx \quad (3)$$

Key **limits*** test

$$\int_{[a,b]} x \, dx \quad (4)$$

$$\int_{[a,b]} \int_{[c,d]} \int_{[e,f]} x \, dx \quad (5)$$

Key **mode** test

$$\int_a^b \int_c^d \int_e^f f^2 g^3 h^4 \, df \, dg \, dh \quad (6)$$

$$\int_a^b f^2 \int_c^d g^3 \int_e^f h^4 \, df \, dg \, dh \quad (7)$$

$$\int_a^b f^2 \, df \int_c^d g^3 \, dg \int_e^f h^4 \, dh \quad (8)$$

Key **symbol** test

$$\iint x \, dx \quad (9)$$

$$\int x \, dx \quad (10)$$

Key **nint** test

$$\iiint x \, dx \quad (11)$$

Keys **llimit** and **ulimit** test

$$\int_X x \, dx \tag{12}$$

$$\int^X x \, dx \tag{13}$$

Key `domain(*)` test

$$\int_{\mathbb{R} \times \mathbb{Z}} x \, dx \tag{14}$$

$$\int_{\mathbb{R}^2 \times \mathbb{Z}^5 \times \mathbb{C}^9} x \, dx \tag{15}$$

$$\int_{\mathbb{R} \times \mathbb{Z}_p} x \, dx \tag{16}$$

$$\int_{\mathbb{E}_t \times \mathbb{A}_t} x \, dx \tag{17}$$

Key `boundary` test

$$\int_{\partial S} x \, dx \tag{18}$$

Key `variables` test

$$\int_a^b \int_a^b \int_a^b xyz \, dx \, dy \, dz \tag{19}$$

$$\int x \tag{20}$$

Key `jacobian` test

$$\int_a^b \int_a^b r^3 \sqrt{\theta} r \, dr \, d\theta \tag{21}$$

$$\int_a^b \int_a^b \int_a^b r^3 \sqrt{\theta} e^\phi r^2 \sin \theta \, dr \, d\theta \, d\phi \tag{22}$$

Key `diff-symb` test

$$\int x \, \mathcal{D}q(t) \tag{23}$$

Key `diff-star` test

$$\int x \, d_x \, d_y \tag{24}$$

Key `diff-options` test

$$\int x \, \mathrm{d}^4 x \quad (25)$$

$$\int_a^b r \, \mathrm{d}^2 (r) \int_a^b \theta \, \mathrm{d}^4 \theta \int_a^b \phi \, \mathrm{d}^6 (\phi) \quad (26)$$

Macro `\NewLimitsKeyword` test

$$\int_{\text{lower}}^{\text{upper}} x \, \mathrm{d} x \quad (27)$$

Macro `\NewVariableKeyword` test

$$\int_a^b \int_a^b \int_a^b x \, \mathrm{d} \alpha \, \mathrm{d} \beta \, \mathrm{d} \gamma \quad (28)$$

Macro `\NewSymbolKeyword`

$$\int_S x \, \mathrm{d} x \quad (29)$$

Macro `\differentials` test

$$\int \frac{\mathrm{d} x}{x} \quad (30)$$

Special syntax test

$$\int_1^{10} x \, \mathrm{d} x \quad (31)$$

$$\int_a^b r r^2 \, \mathrm{d} r \int_a^b \theta \sin \theta \, \mathrm{d} \theta \int_a^b \phi \, \mathrm{d} \phi \quad (32)$$

Messages test

$$\int_a^b x \, \mathrm{d} x \int_a^b \mathrm{d} y \quad (33)$$

$$\int_a^b \int_a^b x \, \mathrm{d} x \, \mathrm{d} y \, \mathrm{d} z \quad (34)$$

$$\int_a^b \frac{1}{x} \int_a^b \frac{\mathrm{d} x \, \mathrm{d} y}{y} \quad (35)$$

$$(36)$$